

# Ocean Guard AI Design Project Description Summary Part II

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## 9 Product Use Cases

Ocean Guard AI has five main use cases that explain how people and system parts work together. These include collecting data from drones, finding predator and prey events, sending alerts, activating safe deterrents, and showing data through charts and maps. Each one shows who is involved, what starts the process, and what happens next.

## 10 Functional Requirements

These requirements describe what the system should do. Ocean Guard AI needs to collect drone data, detect events using AI, send real time alerts, activate deterrents safely, and display data correctly. These functions help the system protect marine life and share useful information with researchers.

## 11 Data Requirements

This section explains what types of information the system collects and stores. It includes drone data, species details, event records, alert logs, and dashboard summaries. This helps make sure all data is clear, organized, and easy to find when needed.

## 12 Performance Requirements

The performance requirements make sure the system works quickly and accurately. Ocean Guard AI must detect events in real time, identify species correctly, and handle data from many drones at once without slowing down.

## 13 Dependability Requirements

Dependability makes sure the system stays reliable and safe. Ocean Guard AI should keep running for long sessions, stay online most of the time, work even if the internet drops, and always activate deterrents safely.

## 14 Maintainability and Supportability Requirements

This section focuses on keeping the system easy to update, fix, and expand. Developers should be able to install updates without stopping operations, users should have access to help and guides, and the system should work for years and support new drones or regions.

## 15 Security Requirements

Security requirements protect system data and user information. Ocean Guard AI must only let authorized users log in, keep private data safe, track all system actions, and block malware or attacks. This helps build trust and keeps the system secure.

## 16 Usability and Humanity Requirements

Usability focuses on making the system easy to learn and use. The interface should be clear and simple so users can react quickly in the field. It should also include options like large text, plain language, color safe design, and multiple languages for different users.

## 17 Look and Feel Requirements

The system should look clean and professional. Ocean Guard AI uses simple fonts, ocean colors, and clear icons so users can focus on their work. The design avoids clutter or distracting effects to make the interface calm and easy to read.

## 18 Operational and Environmental Requirements

Ocean Guard AI must work well in outdoor and marine environments. It should be easy to read in sunlight, work offline when needed, connect to sensors and drones, and be accessible through the web. The system also includes updates to keep it working smoothly.

## 19 Cultural and Political Requirements

These requirements make sure Ocean Guard AI is fair and neutral. The system should use neutral symbols and words, support multiple languages, follow environmental rules, and respect each country's data privacy and policies.

## 20 Legal and Standards Requirements

This section focuses on laws and quality standards. Ocean Guard AI must follow privacy and environmental laws and use secure development practices. It also follows standard data formats so it can share information easily with other research systems.

## 21 Requirements Acceptance Tests

Each requirement connects to one or more acceptance tests that check if the system works as planned. These tests cover everything from drone connection and AI detection to user interface, security, and compliance. They confirm that Ocean Guard AI meets all goals and runs correctly in real use.